

Inside a Computer



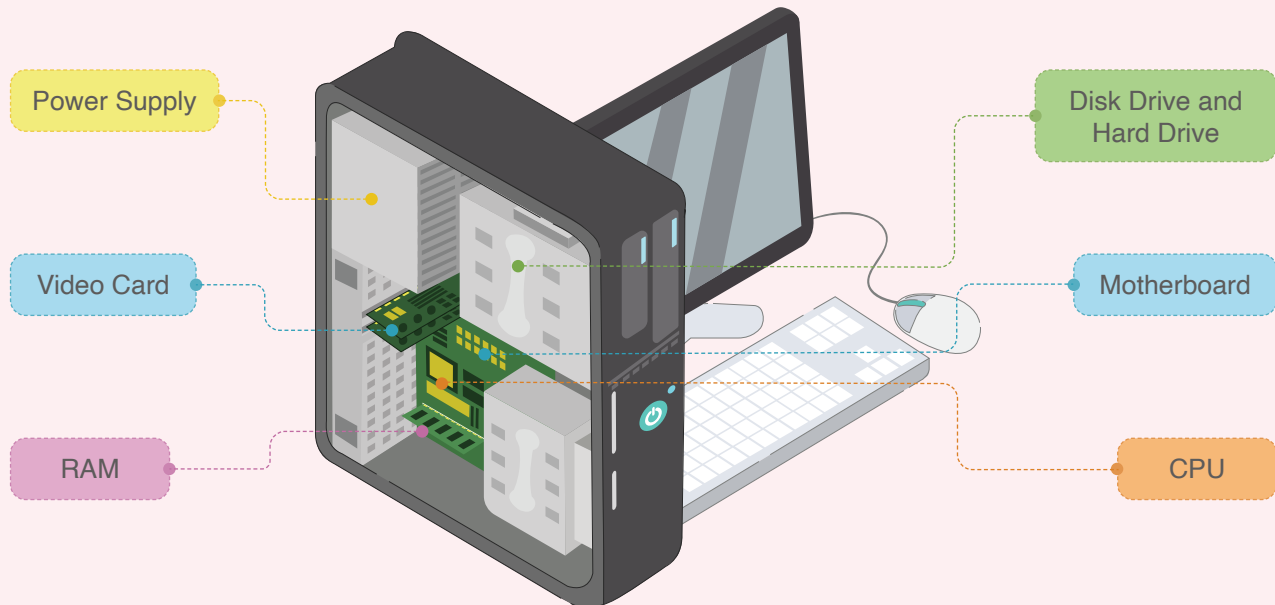
Recall:

A desktop computer has 4 basic components:

1. Monitor
2. Keyboard
3. Mouse
4. CPU Cabinet

1. Inside a Computer

A **CPU Cabinet** is a box that stores all the internal parts of a computer.



Power Supply Unit

- The Power Supply Unit (PSU) provides electric power to the computer. This power is required to turn on and use the computer.



Computer Power Supply Unit (PSU)



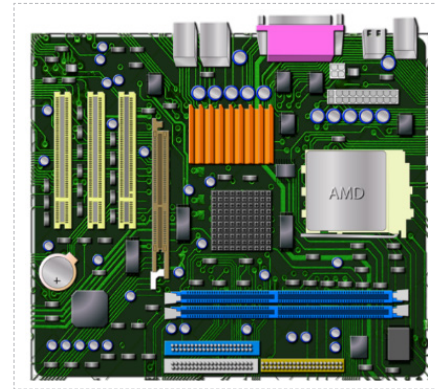
Laptop Power Supply (battery)

A laptop has a battery inside it. This battery works like the power supply unit and provides power to the laptop. With a battery, a laptop can also operate when it is not connected to a power supply.

1. Inside a Computer

Motherboard

- The motherboard is an essential part of the computer.
- It distributes power to other components of the computer.
- It connects all the parts of the computer with each other.



Computer motherboard

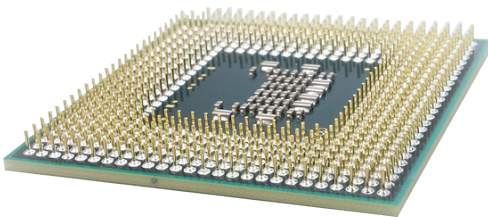
Exercise

Fill in the blanks

- 1 _____ is used to provide power to a computer.
- 2 _____ connects all the parts of a computer together.

Central Processing Unit (CPU)

- The Central Processing Unit is like the brain of a computer.
- The CPU is also called the processor.
- It processes all the information that is sent to the computer.



Central Processing Unit (CPU)

Memory

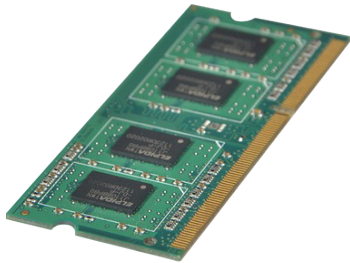
Memory is used by a computer to store data.

Computer memory is of three basic types:

1. Random Access Memory (RAM)
2. Read Only Memory (ROM)
3. Hard Disk Drive (HDD)

Random Access Memory (RAM)

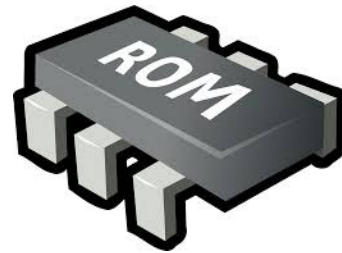
- RAM stores data for a short amount of time.
- RAM is connected to the motherboard inside the CPU cabinet.
- It helps the computer run many software and makes the computer operate faster.



Random Access Memory (RAM)

Read Only Memory (ROM)

- ROM stores data permanently.
- It is a part of the motherboard.
- ROM stores the instructions that are needed to start a computer.



Read Only Memory (ROM)

Hard Disk Drive (HDD)

- The HDD stores data in a computer.
- It is the largest data storage device in the computer.
- All software, files, images & videos can be stored in the HDD.
- HDD is also called hard drive or hard disk.



Hard Disk Drive (HDD)

Bytes

ROM is also used in automatic washing machines & calculators to store instructions.

Exercise

 **Tick Mark ☒ the correct answer**

3 Which of the following stores data for a short period of time in the computer?

a. HDD ☐

b. Power Supply ☐

c. RAM ☐

4 Which of the following is the largest data storage device in a computer?

a. RAM ☐

b. ROM ☐

c. HDD ☐

Video Card

- A Video Card is connected to the motherboard.
- We can see images, videos, and games on the monitor because of the video card.



Video Card

Disk Drive

- The Disk Drive plays CDs and DVDs.
- A disk drive is inside the CPU cabinet or the laptop.
- Some disk drives can be connected externally to the computer.
- A disk drive is also called a CD/DVD drive.



Internal Disk Drive



External Disk Drive



Recall:

CD (Compact Disk): It is a small plastic disk that stores data.

DVD (Digital Versatile Disk): It is a type of CD used to store large amounts of data.

A DVD can store more data than a CD.

Exercise

5 Define Disk Drive.



State whether True or False

6 The PSU allows the processor to send image and video information to the monitor.



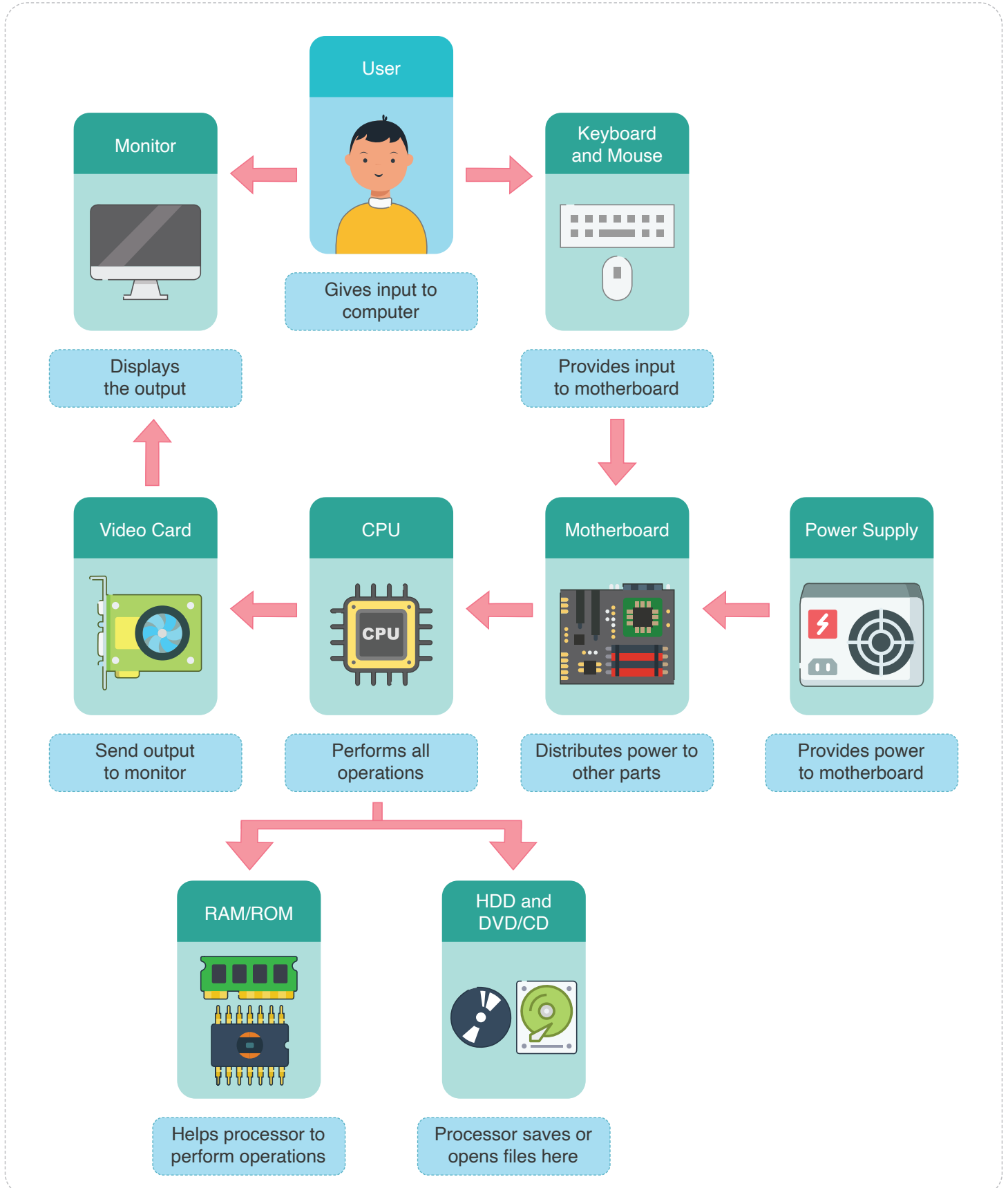
7 A disk drive is also called a DVD drive.



Do you know how a computer works?

- Power supply provides power to the motherboard.
- Motherboard distributes the power to all other parts.
- When we type on a keyboard or click with a mouse, information is sent to the motherboard.
- The motherboard then sends that information to the processor.
- The Processor uses RAM & ROM to perform the operation as per the information.
- The output is then sent to the monitor through the video card.
- At the same time, the processor can save or open the file from a CD, DVD or HDD.

1. Inside a Computer



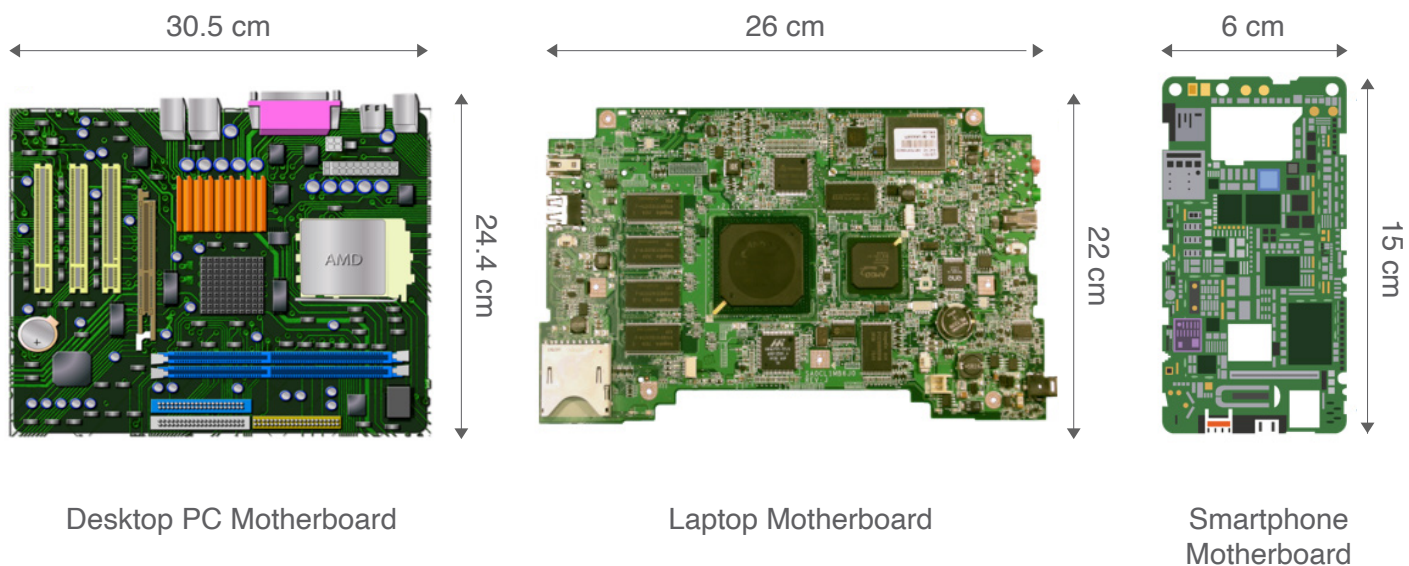
Evolution of computers

With time, new technology has enabled us to evolve our devices to work better and smarter. With this evolution, there have been many changes in computers and their components.

- The size of hardware has reduced. With this, the overall size of a device has reduced too.
- Parts like RAM, Video Card and Hard Disk can be installed separately in the cabinet.
- The size of the PSU is getting smaller and is becoming more portable.
- Different components like the keyboard, mouse, and speakers can be connected via Bluetooth.

Evolution of the Motherboard

Even if the size of the motherboard changes, it plays the same role in all device.



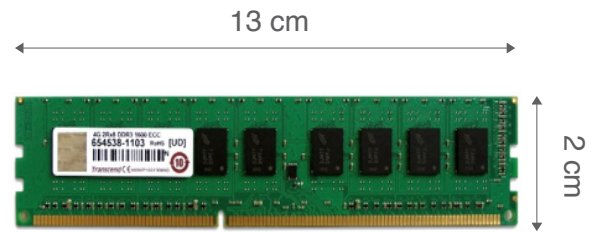
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A machine is considered portable when it can be easily carried and moved to different places.

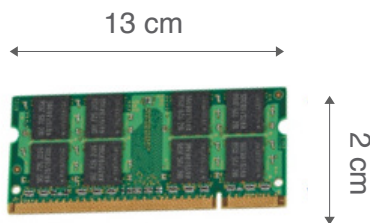
1. Inside a Computer

Evolution of the RAM

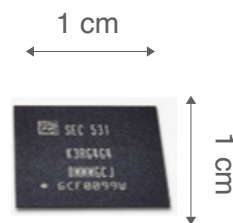
Even if the size of the RAM changes, it temporarily stores data in all devices.



Desktop PC RAM



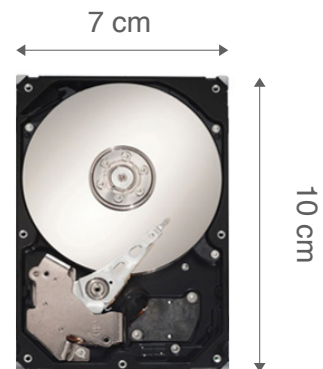
Laptop RAM



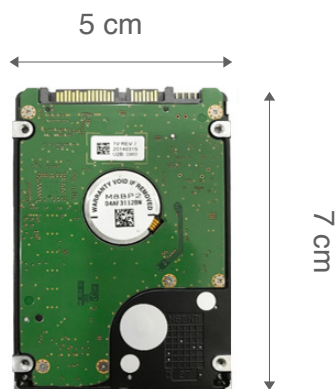
Smartphone RAM

Evolution of the Hard Disk

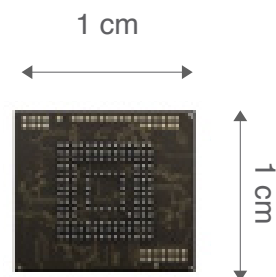
Even if the size of the hard disk changes, it permanently stores data in all devices.



Desktop PC Hard Disk



Laptop Hard Disk

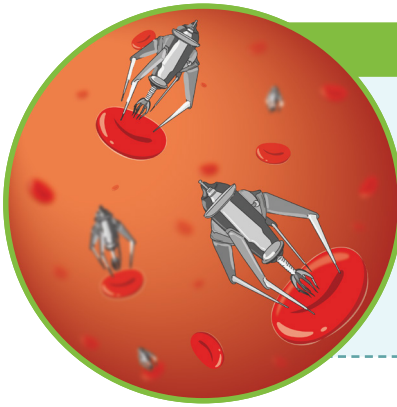


Smartphone RAM



Tech Flash

- CPU** : CPU is the brain of the computer. All the instructions given by the user are performed here.
- Memory** : Memory is used by computers to store all the data.
- RAM** : RAM is used by the computer to run all software.
- Video Card** : The processor uses the video card to send information of the image, video and game to the monitor.
- Disk Drive** : Disk drive is the device that is used to read or write the data stored in CDs or DVDs.



Future Tech

Nano-Computer

A Nano computer is so small in size that it can only be seen through a microscope. Nano computers are similar to robots. They are being developed mainly to work in tiny places like blood vessels.

